

Subject: Mandatory Recalibration of Model Evaluation Standards: A Critique of Current Practices and Directive for Clinical Validity

To: Dr. Chloe Bennett, Computational Lead; Bob Richardson, Machine Learning Engineer; All Members of the Modeling and Validation Teams

Cc: Dr. Aris Thorne, Clinical Neurology Lead; Statistical Review Board

This memorandum addresses a fundamental and concerning deficiency in the methodological rigor of our internal model evaluations, as evidenced in the recent Q4 2025 progress report (Ref: AIAD-ML-Report-Q4-2025_v12). The continued primary reliance on "Simple Accuracy" as a key performance indicator for our early-stage Alzheimer's disease (AD) classification tasks is not only statistically naïve but clinically dangerous. It reflects a profound misunderstanding of the translational context of our work and threatens to invalidate months of research effort. This practice must cease immediately.

Critique of the "Simple Accuracy" Fallacy

A reported aggregate accuracy of 90.2% for distinguishing prodromal AD (MCI converters) from stable MCI (non-converters) is, in isolation, a meaningless—and potentially misleading—figure. In a clinical population where the prior probability of conversion is not 50%, and where the cost of errors is profoundly asymmetric, accuracy is an invalid metric.

Consider a hypothetical cohort with a 15% conversion rate. A model that simply labels every single case as "non-converter" would achieve an accuracy of 85%. Our reported 90.2% suggests only marginal improvement over this clinically useless baseline. The critical failure of your report is the obfuscation of error distribution.

The Imperative of Error Asymmetry in Early Diagnosis

For a progressive, irreversible condition like Alzheimer's disease:

- A False Positive (Type I Error) may cause undue psychological distress and lead to unnecessary follow-up investigations, a significant cost.
- A False Negative (Type II Error) is catastrophic. It represents a missed opportunity for early intervention, patient and family planning, and potential enrollment in clinical trials at the most efficacious stage. A model with 90% accuracy but a 25% False Negative Rate is ethically and clinically untenable. It would fail in any real-world deployment, regardless of its impressive-sounding headline accuracy.

Directive for New Evaluation Standards

Effective immediately, all future internal model validation reports, publications, and presentations must depose "Accuracy" as a primary metric. The following suite of metrics, which account for class imbalance and error cost, is mandatory:

1. Confusion Matrix with Prevalence-Adjusted Metrics: All reports must lead with the full confusion matrix, from which the following must be calculated and emphasized:
 - Sensitivity (Recall/True Positive Rate): The proportion of actual converters we correctly identify. This is our most critical metric.
 - Specificity (True Negative Rate): The proportion of stable cases correctly identified.
 - Precision (Positive Predictive Value): Of all cases predicted to convert, what proportion truly did? This speaks to the clinical utility of a positive prediction.
2. The Precision-Recall (PR) Curve: The Receiver Operating Characteristic (ROC-AUC) curve, while provided, can be overly optimistic with imbalanced data. The PR curve is mandatory for all classification tasks. The Area Under

the PR Curve (AUPRC) provides a far more realistic assessment of model performance on our minority class (converters).

3. The F1-Score (and $F\beta$ -Score): The harmonic mean of precision and recall. I further direct the team to explore the $F\beta$ -Score, where β can be weighted ($\beta > 1$) to emphasize recall over precision, reflecting our clinical priority to minimize False Negatives.
4. Cohen's Kappa Coefficient: This metric measures the agreement between the model and ground truth, corrected for chance agreement expected from the class distribution. A high accuracy with a low Kappa (as we likely have) explicitly reveals that the model is leveraging the imbalance, not learning robust discriminative features.

Mandatory External Validation Protocol

The report's validation on a held-out test set from our internal Knight ADRC cohort is necessary but insufficient. It does not account for center-specific biases, notably the hardware acquisition protocols of our MRI scanners. A model may be learning "scanner fingerprint" artefacts rather than true neuropathological signatures.

Therefore, I mandate a new, non-negotiable step in our validation pipeline:

- All finalized models must be evaluated on the OASIS-3 dataset. This large, open-access, multi-scanner, longitudinal neuroimaging dataset provides the necessary heterogeneity to stress-test the generalizability of our models. Performance degradation on OASIS-3 will be a key measure of overfitting and will guide our feature engineering and regularization strategies. A model that performs well only on our local data is a scientific dead-end and will be unpublishable in a high-impact journal.

Conclusion and Action Items

The sophistication of our neural network architectures must be matched by the sophistication of our evaluation frameworks. To submit a manuscript to *Nature Medicine* boasting "90% accuracy" would invite immediate desk rejection and professional ridicule from the clinical neuroinformatics community.

1. Re-analysis: Chloe and Bob, you are to re-analyze all current candidate models using the above metric suite. The revised report is due by 2026-01-19.
2. Protocol Update: By 2026-01-26, you will submit a revised Model Validation Protocol document incorporating these standards and the OASIS-3 validation step for my signature.
3. Training: A seminar for all project personnel on clinically meaningful ML evaluation will be scheduled before the end of January, led by the Statistical Review Board.

We are building diagnostic tools, not academic curiosities. Our statistical rigor is the foundation of our clinical validity and our ethical responsibility. There will be no compromise on this front.

Dr. Alice Robertson, Ph.D., F.R.S.M.

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